UCLA
Dept. of Electrical and Computer Engineering
ECE 214B: Advanced Topics in Speech Processing
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Mini Project 1: Fine-Tuning ASR Models for Child Speech

Due: 4/27/2026
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I. Introduction

This project focuses on fine-tuning Automatic Speech Recognition (ASR) models for child speech. Child speech presents unique challenges due to differences in pitch, pronunciation, and variability compared to adult speech. The goal is to adapt pretrained ASR models for improved performance on child speech datasets using lightweight fine-tuning strategies.

II. Dataset

The CSLU OGI Kids [1] corpus contains around 50 hours of speech from 1100 children (from kindergarten to grade 10) consisting of both read and spontaneous speech. This project uses the read-speech portion of the corpus, where utterances consist of simple words, sentences, or digit strings. The version of the dataset provided to you is a subset of the original dataset to enable easier execution on Google Colab. The provided subset contains 20,000 training samples, 2,000 validation samples, and 2,000 test samples

III. Project Codebase

The enclosed project package contains two folders: OGI_Kids and data. The OGI_Kids folder contains all the necessary audio files for the project. The data folder contains the reference files for indexing the train, validation, and test sets for the task, along with the transcriptions for the files. The accompanying python notebook [2] contains a pipeline to fine-tune the ASR models based on the data provided in the index files.

IV. Objectives

The primary objective is to fine-tune two pretrained ASR models: Whisper [3] and Wav2vec 2.0 [4] to optimize performance on child speech. Wav2vec 2.0 is pretrained on LibriSpeech, while Whisper is pretrained on a web-scale proprietary dataset. This includes exploring the performance of different models in zero-shot (aka no fine-tuning) and fully fine-tuned settings. The project is structured into the following tasks:

1. Experiment with Whisper-base.en (74M) and Whisper-tiny.en (39M) models and compare the performance in zero-shot and full fine-tuning settings.

2. Experiment with Wav2vec 2.0 base (95M) and compare the performance in zero-shot and full

fine-tuning settings.

3. For the remaining experiments, use only the base models.

   Explore the impact of learning rates (how much the model adjusts its weights), batch sizes (number of samples per model update), and max_steps (total number of updates) on reducing Word Error Rate (WER).

4. Using the best configurations from (3), examine the effect of training dataset size on recognition accuracy on the test set. For example, decrease the number of training samples to 10000 and 5000.

5. Analyze WER differences across age groups (4–14). Experiment with models fine-tuned on samples from specific age ranges (4-6, 7-9, 10-14, all ages) by testing on samples from other age ranges.

6. **Bonus:** Examine the effect of data augmentation [8][9][10] on system performance.

V. Instructions

A. Download project package from Box [5].
B. Unzip the compressed file.
C. Upload the 'OGI_Kids' and 'data' folders to your Google Drive account
D. Make a copy of the Colab notebook [2]
E. Open the notebook and run all the cells

VI. Baseline Results

The baseline script should take around 3 hours to run on Google Colab.

Wav2vec 2.0 base	**Word Error Rate (WER)**	**Whisper-base.en**	**Word Error Rate (WER)**
Zero Shot	**48.2%**	**Zero Shot**	**28.5%**
Fully Fine-tuned	**31.7%**	**Fully Fine-tuned**	**27.1%**

VII. Oral Presentations

There will be oral presentations by the different teams describing their work. Presentations should be planned by the team as a group.

VIII. Report and Code

The report (one per group) should include:

- Introduction (what is the problem/why is it important)
- Background (literature survey)

● Project Description (algorithm implementation, results, average run times, etc.)
● Summary and Discussion (also ideas for future work)
● References (cited throughout the report)
● Figures and flowcharts generally help clarify the text.

The report should be 4-pages long (excluding references) and have the same format as the INTERSPEECH conference. The presentations will be on **4/27** and the final reports may be turned in on **4/29.**

Useful References

1. CSLU OGI Kids Dataset:
   Shobaki, Khaldoun, John-Paul Hosom, and Ronald Cole. "The OGI kids' speech corpus and recognizers." Proc. of ICSLP. 2000.

2. Google Colab Notebook for feature extraction and classification:
   https://colab.research.google.com/drive/1-XNMsTVWxQ0J0878JU8I5salkmNZL7WT?usp=sharing

3. Whisper family of models
   Radford, Alec, et al. "Robust speech recognition via large-scale weak supervision." International conference on machine learning. PMLR, 2023.

4. Wav2vec2.0
   Baevski, Alexei, et al. "wav2vec 2.0: A framework for self-supervised learning of speech representations." Advances in neural information processing systems 33 (2020): 12449-12460.

5. Package with all training and test files for the project:
   https://ucla.box.com/s/dkhdvyd4qpmp29ls1vim3z3lfauy5hz1

6. Transformers library
   Wolf, Thomas, et al. "Transformers: State-of-the-Art Natural Language Processing." EMNLP 2020 (2020): 38.

7. Fan, R., et al. (2024) Benchmarking Children's ASR with Supervised and Self-supervised Speech Foundation Models. Proc. Interspeech 2024.

8. Patel, Rupal, et al. "Prosodic adaptations to pitch perturbation in running speech." Journal of Speech, Language, and Hearing Research 54.4 (2011): 1051-1059.

9. Jaitly, Navdeep, and Geoffrey E. Hinton. "Vocal tract length perturbation (VTLP) improves speech recognition." Proc. ICML workshop on deep learning for audio, speech and language. Vol. 117. 2013.

10. Ko, Tom, et al. "Audio augmentation for speech recognition." Interspeech. Vol. 2015. 2015.